Night shift — matched COBA and PING on SHD

ar066 · 18 July 2026 · Draft

Digest

This section is intentionally empty until the shift finishes. It will report what completed, what failed its registered criterion, the compute used, and the next question worth testing.

Mandate

Programme goal

Establish a minimal working Spiking Heidelberg Digits (SHD) baseline for matched COBA and PING networks under Dale's law. Both networks must learn above chance on a small subset, produce stable learning curves, and expose input, excitatory, and inhibitory spike rasters that can be compared on the same utterances. This shift establishes feasibility. It does not test whether PING is more accurate, more efficient, or better able to use temporal structure than COBA.

Why this is the first question

The existing MNIST training hub shows that both modes of the same conductance-based network learn a static image task. COBA disables the reciprocal excitatory-to-inhibitory-to-excitatory loop, while PING keeps the fixed loop and produces gamma-rhythmic population activity. SHD is the next test because its input already consists of timed cochlear spikes rather than an image converted into Poisson events. Before comparing any computational advantage, the two established modes must first train cleanly on this input and yield interpretable dynamics.

Registered experiment

Field	exp066
Hypothesis	With matched input and readout settings, both Dale-constrained COBA and PING learn a small SHD classification task above chance and produce stable, directly comparable population rasters.
Baseline	The 5% chance floor for SHD's 20 classes; both models must exceed 20% test accuracy in the pilot.
Kill criterion	Stop before scaling if either model remains below 20% test accuracy, develops a non-finite loss, or stays silent or saturated after the single pre-registered shared recipe adjustment.
Seed count	1

Field	exp066
Budget	One local smoke stage plus at most two RunPod training attempts per model; 2 h per attempt and $40 total RunPod spend.
Status	running

The comparison has two predeclared architectural differences. COBA disables the inhibitory loop and uses no voltage-gradient dampening. PING enables the fixed inhibitory loop and uses the established voltage-gradient dampening factor of 1000. The dampening is part of PING's training recipe and will not be tuned after comparing outcomes.

Everything else is matched: one hidden layer with 256 excitatory cells and 64 inhibitory cells, Dale's law, fixed recurrent weights, identical input weight initialization and sparsity, identical readout initialization and mode, Adam learning rate, batch size, integration timestep, simulation duration, data subset, epoch count, and seed. The proposed first recipe uses a shared input-weight mean of 0.9, 95% input sparsity, a readout scale of 225, membrane-mean readout, learning rate 0.0004, batch size 32, integration timestep 1 ms, simulation duration 1000 ms, seed 42, and no firing-rate regulariser.

Staged execution

1. **Smoke stage.** Train both cells on 128 training utterances for 2 epochs on the local CPU. This stage checks the SHD loader, tensor shapes, finite loss, non-silent hidden activity, checkpointing, and recording. It is plumbing, not evidence for the hypothesis.
2. **Pilot stage.** If both smoke cells are finite and active, train both cells on the same 1000-utterance subset for 20 epochs on one RunPod GPU. The pilot is the registered result.
3. **Shared adjustment.** If either smoke cell is silent or saturated, both models receive one identical change to the input-weight scale before the smoke stage is repeated. No model-specific input or readout tuning is permitted. If the repeated smoke still fails, the experiment is killed and the pilot does not run.

Evidence packet

The experiment must report train and test loss by epoch, test accuracy by epoch, excitatory and inhibitory firing rates, non-finite losses or skipped updates, and the exact configuration. It must also render input, excitatory, and inhibitory rasters for the same pre-selected test utterances in both models. The utterances are selected by dataset position before predictions are inspected, and every panel uses the same time axis and population order.

Operating contract

```
budgets:
  wall_clock_per_run: 2h
  wall_clock_per_shift: 8h
  runpod_total_usd: 40
  max_pods_concurrent: 1
  seeds_default: 1
scope:
  collection: spiking-heidelberg-digits
  may_propose: true
  may_edit_snn_cli: false
stop_when:
  - queue_empty
  - shift_budget_exhausted
  - runpod_budget_exhausted
  - build_red_twice
```

Branch and logging contract

This programme runs on the existing `autoresearch/shd` pull-request branch. The mandate is committed there before the experiment runner or any result, making that commit the pre-run anchor. The branch is merged into `main` only if the scientist accepts the completed evidence packet.

Implementation changes, debugging decisions, and other engineering work are recorded in focused commit messages so the pull request carries a readable code history. Scientific events and conclusions are recorded here in the Record as the shift proceeds. The Record may link to a commit when the exact implementation change is relevant to interpreting a result, but it does not duplicate the engineering log.

Record

Local smoke stage

The registered seed-42 smoke passed for both cells at the original shared input-weight scale, so the pre-registered shared adjustment was not used. Both losses and all recorded diagnostics remained finite, neither optimiser skipped an update, and both hidden populations were active. The COBA cell had excitatory activity with its disabled inhibitory loop, while the PING cell had both excitatory and inhibitory activity. The local accuracy values are plumbing diagnostics only and are not evidence for the hypothesis.

The first implementation attempt incorrectly equated the fraction of cells that fired at least once with saturation. Inspection showed that the COBA population had a modest firing rate even though every cell fired during the one-second utterance. The gate was corrected to use near-timestep-rate firing as saturation, matching the meaning of the registered criterion. This did not alter a scientific setting or require a rerun. The runner and corrected gate are recorded in commit e42b264.

Pilot dispatch attempts

The first pilot dispatch created no pod because the local dispatch host did not have the deployment public key expected by the existing RunPod helper. No model ran and no RunPod charge accrued. This was an infrastructure abort, not an observation of either registered cell.